

**Source**  
Seed: [liambee.me/.../troubleshooting-plc-systems-a-systematic-approach-to-finding-faults](https://liambee.me/.../troubleshooting-plc-systems-a-systematic-approach-to-finding-faults)  
Rival claims (4): [automate.org/.../troubleshooting-tips-isolating-intermittent-faults](https://automate.org/.../troubleshooting-tips-isolating-intermittent-faults)  
Rival claim (1): [ecsintl.com/troubleshooting-intermittent-plc-faults-a-guide-to-...](https://ecsintl.com/troubleshooting-intermittent-plc-faults-a-guide-to-...)

**Verbatim span (seed)**  
“Loose terminals cause many intermittent faults. A flashlight reveals problems in dark panels that you miss otherwise.”

**Claim ID + label**  
c-d7bce12d16755797 — literature\_supported — verdict supports  
Assertion (the extractor’s paraphrase, which the lens reads): “Loose terminal connections are a common cause of intermittent PLC faults and should be checked during troubleshooting.”  
*The anchor is this assertion cut at 80 characters (at a word boundary): “Loose terminal connections are a common cause of intermittent PLC faults and”.*

**Lens fired: DIAG (rival causes)**  
7 independent sources discuss the topic; the lens found 5 rival causes (listed in the hypothesis below) and looked for a discriminating sign for each.

**Inferred gap — closure\_state: open**  
Closure judge (qwen2.5:7b-instruct, local), 34 verified sentences retrieved over the 5 rival probes: no rival fully distinguished; 4 of 5 partly addressed, with 5 partial rival hits; the fifth (aliasing) matched only an unverified sentence. The DIAG rule counts a partial rival as unsigned, so the group reads open. Judge, verbatim: “Sentences 1 and 4 partially address the distinction but do not explicitly state an observable sign.” Re-runnable: prompt sha 9c50fb9250a8.

**Hidden-knowledge hypothesis (label: inferred), verbatim**  
“Hypothesis (inferred): practitioners are predicted to tell these rival causes of “Loose terminal connections are a common cause of intermittent PLC faults and” apart by signs; the closure judge found a sign for the following (test DIAG:c-d7bce12d16755797):  
- Ground loops and interactions of multiple processors finding weaknesses in (no sign found)  
- Aliasing in data can be a (no sign found)  
- by installation problems, while software problems are typically less (no sign found)  
- by high-frequency electrical noise from VFDs and contactors that creates phantom (no sign found)  
- Improper pull-up resistors and incorrect clock edge timing in handshake lines (no sign found)”  
*The template prints “(no sign found)” for every rival not fully signed, including the 4 partly signed ones.*

**Template question + channel, verbatim**  
“Sources list several causes of Loose terminal connections are a common cause of intermittent PLC faults and: “Next are unwanted interactions, be they ground loops or interactions of multiple processors finding weaknesses in the hardware or software interfaces.”, “high-frequency noise from VFDs and contactors creates the “phantom faults” that disrupt your logic signals and ruin your production schedule.”. Think of the last time you diagnosed Loose terminal connections are a common cause of intermittent PLC faults and. Which cause did you suspect first, and what did you see, hear or measure that let you rule the others out?”  
Channel: CDM probes on a recalled case, then contrasting cases

**Expert answer**  
Not collected: zero participants by design (NOW program).